

SANJAYKUMAR M

AI ENGINEER · MACHINE LEARNING · COMPUTER VISION

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PROFESSIONAL SUMMARY

Aspiring AI Engineer graduating in 2026 with hands-on experience in Machine Learning, Deep Learning, Computer Vision, NLP, and Generative AI. Developed AI-powered solutions including a DistilBERT-based sentiment analysis platform (92.4% accuracy), a cloud-native multi-agent RAG platform, and an oil spill detection system using satellite imagery and AIS data. Proficient in Python, PyTorch, Hugging Face Transformers, OpenCV, FastAPI, LangChain, and LangGraph. Smart India Hackathon 2024 Grand Finalist and 3rd Prize winner at GALAXY'24 Phase-II. Eager to contribute to innovative AI teams and build scalable, production-ready intelligent systems.

TECHNICAL SKILLS

Machine Learning & AI: Deep Learning, Machine Learning, Natural Language Processing (NLP), Computer Vision, Generative AI, Sentiment Analysis, Anomaly Detection

Frameworks & Libraries: PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, MediaPipe, Streamlit, Matplotlib

Databases & Tools: MySQL, ChromaDB, Jupyter Notebook, VS Code, Antigravity IDE

Generative AI & LLMs: RAG Pipelines, LangGraph, LangChain, Hugging Face Transformers, Llama 3, DistilBERT, Prompt Engineering, LLM API Integration, Groq API

MLOps & Infrastructure: FastAPI, GitHub, Docker, REST APIs, Model Serving, CUDA

Programming Languages: Python, SQL, C

PROJECTS

Oil Spill Detection

- ▶ Built a machine learning model integrating AIS data and satellite imagery to detect marine oil spills.
- ▶ Implemented anomaly detection techniques to identify suspicious vessel behavior and potential spill events.
- ▶ Enhanced environmental monitoring by enabling early detection and analysis of maritime pollution incidents.

Key Technologies: Python, Machine Learning, Computer Vision, Satellite Imagery, AIS Data Analysis

AI Virtual Painting

- ▶ Created a real-time gesture-controlled virtual painting application using hand tracking and computer vision.
- ▶ Implemented EMA smoothing, interpolation, and gesture debouncing for accurate and stable drawing.
- ▶ Designed an interactive web dashboard with real-time synchronization, undo/erase tools, and artwork export functionality.

Key Technologies: Python, OpenCV, MediaPipe, Computer Vision, Hand Tracking

Sentiment Analysis Dashboard

- Engineered a DistilBERT-based sentiment analysis platform achieving 92.4% accuracy for real-time sentiment prediction.
- Built a scalable AI pipeline using Hugging Face, PyTorch, and FastAPI for efficient model inference and deployment.
- Integrated SHAP Explainable AI (XAI) and optimized model training using FP16 mixed precision and CUDA acceleration.

Key Technologies: Python, PyTorch, Hugging Face Transformers, DistilBERT, NLP, FastAPI, SHAP, Streamlit, CUDA, Machine Learning

Cloud-Native Multi-Agent RAG Platform

- Designed a multi-agent RAG system using LangGraph, ChromaDB, and Llama 3 for intelligent document retrieval and reasoning.
- Engineered semantic chunking, knowledge graph extraction, and citation-based retrieval to improve answer accuracy and explainability.
- Integrated Groq-powered LLM inference and built an interactive Streamlit interface for real-time knowledge exploration.

Key Technologies: Python, LangGraph, LangChain, Groq API, Llama 3, ChromaDB, RAG, Knowledge Graphs, Streamlit, Generative AI, Vector Databases

EDUCATION

Bannari Amman Institute of Technology

Graduated: 2026

Bachelor of Engineering in Computer Science Engineering

CGPA: 7.06 / 10.0

G.H.S.S School

Graduated: 2022

Higher Secondary (12th Grade): 79.16% | Secondary (10th Grade): 71.8%

INTERNSHIP

SourceSys Technologies Private Limited

Feb 2026 – Apr 2026

- Successfully completed a Data Science internship, gaining practical experience in data analysis and machine learning.
- Worked with Python and data science tools to analyze datasets and support data-driven solutions.

AWARDS & ACHIEVEMENTS

- ★ **Finalist, Smart India Hackathon (SIH) 2024** — Recognized by the Ministry of Earth Sciences for innovative machine learning applications in marine conservation. Competed among top 50 teams out of 10,000+ registrations nationwide.
- ★ **3rd Prize, GALAXY '24 Phase-II** — Awarded at a National Level Technical Symposium for a comprehensive research presentation on AI Virtual Painting frameworks and real-time computer vision systems.